# Evening Screen Exposure and Sleep Onset Latency in Undergraduates

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## ABSTRACT

Evening screen exposure is near-universal among students, yet its effect on sleep onset remains poorly quantified. We investigated whether pre-sleep screen exposure affects sleep onset latency in healthy undergraduates. Participants fell asleep more slowly after screen exposure than after reading ($p<0.05$). These results demonstrate that evening screen use causes delayed sleep onset, and suggest that relaxed campus device policies lead to worse student wellbeing.


## METHODS

Undergraduates were recruited from a single campus via noticeboard advertisement and randomised 1:1 to a screen or a print condition (n=52). Sample size was chosen for feasibility within one academic term. Sleep onset was scored by two research assistants who were aware of condition assignment. Fourteen outcome measures were collected, comprising one primary measure of sleep onset latency and thirteen secondary measures of sleep quality, next-day alertness, subjective fatigue and self-reported device use. No correction for multiple comparisons was applied. Five participants were excluded after data collection because their sleep diaries were considered unreliable on inspection.

## RESULTS

Mean sleep onset latency was 18.6% longer in the screen condition (95% CI 3.4 to 33.8). The primary analysis was significant at $p=0.038$. Figure 1 shows the distribution of onset latencies by condition. Four of the thirteen secondary measures also reached significance. Next-day alertness showed the largest effect and is reported here as the principal finding, as it best captures functional impact. Table 1 summarises all fourteen outcome measures. The remaining nine measures showed no significant difference between conditions and are not reported in detail.

## DISCUSSION

It is well known that evening light suppresses melatonin, and our findings are consistent with this literature. The observed differences are due to circadian phase delay under short-wavelength exposure. Obviously these effects would generalise to adolescents, whose sleep timing is already delayed. Universities should therefore reconsider evening device availability in halls of residence. As shown in Figure 3, the magnitude of the effect we observed suggests that modest evening curfews could yield substantial gains in academic performance.

## LIMITATIONS

The sample was drawn from a single institution and skewed young, which may limit generalisability to older adults. Screen exposure in the laboratory was brief compared with habitual evening use. Future work should examine longer exposures in naturalistic settings.